

Instructions: Solve all questions sequentially. Illustrate your answers with neat sketches, diagrams wherever necessary. If any data is missing, assume suitable data giving reason.

Q.No

- 1 (a) Explain the need for energy sources. With their merit and demerit enlist different types of conventional energy sources. [5]
(b) Enlist different non-conventional energy resources. Explain briefly constraints in use of these resources. [5]
2 (a) Explain principle of heat generation in resistance heating. Discuss on factors which affect the efficiency of indirect resistance heating method. [6]

OR

- (a) With necessary diagram explain principle of dielectric heating. Discuss on merits and applications. [6]
(b) With neat figure explain functioning of Ajaxwayatt type induction furnace [3]
(c) Calculate the efficiency of a high frequency induction furnace which takes 10 minutes to melt 1.8 kg of aluminum. The input to the furnace being 4.8 kW and initial temperature is 15°C. Specific heat of Al=0.88kJ/kg°C, melting point of Al=660°C; Latent heat of Al=32kJ/kg; 1kJ=2.78 X10⁻⁴ kWh. [3]
3 (a) Enlist different electric loads used in commercial sector. State their specifications. With neat diagram explain the operation of HV AC system used in typical commercial establishment. [6]

OR

- (a) Explain the use of electrical energy in municipal sector. Enlist different electric loads used in municipal sector. With neat block diagram explain the operation of energy efficient street lighting system. <https://www.batuonline.com> [6]
(b) With neat diagram explain the use of electrical energy in agriculture sector. Enlist their requirements [3]
(c) Write a brief note on infrared heating. [3]
4 (a) Explain briefly need for energy audit in an industry. Enlist its classification. State their merits and demerits. [4]

OR

- (a) Define Sankey diagram. Explain how it is helpful in energy audit study. [4]
(b) Explain with suitable example ABC analysis used in energy audit. [4]
(c) In an Industry it was observed that the different phases are loaded with 100A, 200A and 150 A. Explain how energy audit will lead to energy conservation in that industry? [4]
5 (a) Differentiate clearly between supply side management and demand side management. Enlist

different types of electric tariff. Explain how differential tariff helps in DSM? [6]

OR

- (a) Explain briefly scope for energy conservation in a transmission line. [6]
- (b) Derive expression for optimum value of power factor; hence determine new value of power factor when the tariff is Rs 180/ KVA of maximum demand plus a flat rate/kwh. Assume additional cost of condenser as Rs 140/ KVAR. ROI and ROD together taken as 10% [6]
- 6 (a) With suitable data discuss on energy conservation opportunities in a sugar industry [4]
- (b) Two systems of tariff are available for a factory working 8 hours a day for 300 working hours in a year.
- a) H.V. Supply at a 2.5 Rs/unit plus 250 Rs/kVA of maximum demand.
- b) L.V. Supply at 500 Rs/ kVA plus 4.5 Rs./unit. The factory has an average load of 200 kW at a p.f of 0.95 lag and maximum demand of 250 kW at the same p.f. The H.V equipment cost Rs 150/kVA and losses can be taken as 4.5%. Assume ROI and depreciation to be 12% . Calculate best option and comment on your answer. [4]
- (c) Enlist different methods of power factor improvement in industrial sector. Explain static capacitor method. [4]

OR

- (c) Define payback period. With suitable example explain method of calculation of payback period. [4]

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